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Predicting Hepatocellular Carcinoma Graft Survival Rate in Post Liver Transplantation Using DeepHit

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Abstract

Hepatocellular carcinoma (HCC) is still a severe worldwide health issue, and liver transplantation is frequently the primary curative option for people who meet the criteria for it. It is challenging to predict the long-term graft survival of HCC patients after transplantation. However, it is an essential clinical decision-making task. The DeepHit algorithm is used in this study to predict the long-term survival of grafts using a deep learning-based approach. In this study, we used the dataset collected from the Amrita Institute of Medical Science & Research Centre, Kochi, with patient-specific information on age, gender, tumor size, AFP levels, cirrhosis status, treatment approach, recurrence, comorbidity, and time to transplant failure. The temporal correlations in patient data are modeled, and censoring is considered using the DeepHit technique. It blends survival analysis with the benefits of deep neural networks. DeepHit predicts long-term graft survival in liver cancer patients after transplantation with 94 % accuracy, outperforming conventional methods and improving prognosis and quality of life.

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Keywords: DeepHit; Hepatocellular carcinoma; Graft survival analysis; Post-liver transplantation

1. Introduction

Hepatocellular carcinoma (HCC) patients have a crucial and frequently life-saving therapeutic option in liver transplantation. HCC, the most fatal primary liver cancer, has a convoluted etiology. There is a high likelihood of it coexisting with other chronic liver conditions such as cirrhosis, non-alcoholic fatty liver disease, and hepatitis B or C infections. HCC has a tendency for aggressive growth and metastasis, one of its distinguishing characteristics and making it a strong foe in oncology [1]. Due to its exceptional ability, not only to eradicate the cancerous tumor but

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also to restore a patient to health and normal liver function, liver transplantation assumes essential importance in this situation[2]. By treating both the malignant tumor and the underlying liver condition, this surgical procedure provides a complete remedy, giving the patients a new life.

The ability of liver transplantation to remove cancer reduces the chance of tumor recurrence, which reveals the procedure's tremendous benefits [3]. Beyond this, liver transplantation involves substituting a healthy donor liver for a damaged liver, which may potentially stop the advancement of underlying liver diseases like cirrhosis and aid in recovering vital liver functions like detoxification, metabolism, and protein synthesis [4]. In addition to prolonging the lives of carefully chosen HCC patients getting organs from compatible donors, this revolutionary method also markedly improves their overall quality of life [5]. Hepatocellular carcinoma is a significant public health concern in the global liver cancer landscape, frequently requiring liver transplantation for effective therapy [6]. This therapeutic opportunity, providing an uncommon potential for a cure, necessitates an expedient diagnosis, efficacious treatment, and efficient distribution of organs [7]. Although the goal of liver transplantation is to replace a diseased liver with a healthy one in order to extend a patient's life and restore liver function, patient selection remains challenging because of things like strict criteria and limited availability of healthy donors [8]. Especially when it comes to liver transplantation for patients with HCC, predictive models and algorithms are crucial [9].

HCC patients undergoing liver transplants, the DeepHit technique, a machine-learning framework intended for time-to-event forecasting, appears as a ray of hope. The DeepHit algorithm, a predictive modelling tool in the healthcare industry, is a significant player in predicting graft survival for patients with hepatic cancer following liver transplantation, thanks to its capacity to identify complex relationships in data [10]. This novel method integrates patient-specific information, including clinical, pathological, and demographic data, to improve predictive accuracy and guide therapeutic decision-making in this crucial area [11].

1.1. Objectives of the Paper

The research mainly focuses on developing a forecasting model that uses the DeepHit algorithm to estimate the probability that patients with HCC undergoing liver transplants will have long-term graft survival. However, the DeepHit algorithm can capture complex and time-varying correlations across clinical, pathological, and demographic factors, and outperform traditional predictive models in predicting the possibility of graft survival over an extended period of time. In the long run, achieving this goal will improve long-term outcomes and resource allocation in HCC liver transplantation by providing physicians with a reliable and adaptable tool for patient risk assessment and post-transplant care optimization.

1.2. Organization of Paper

Other sections of the paper are organised as follows: In-depth reviews of prior research works is provided in Section 2. Materials and methods used in the study are discussed in Section 3. The section 4 highlights the proposed methodology. In Section 5, we present the results and analysis of our research work.Finally, section 6 concludes the paper with future directions.

2. Related Works

Liver transplantation is crucial for Hepatocellular Carcinoma patients, and accurate graft survival prediction is crucial for clinical decision-making. Deep learning, in particular, can now be used by patients with HCC to predict graft survival. This study discusses the development of prediction models, their advantages and disadvantages, and the DeepHit algorithm's promise for improving transplant survival prediction.

Lekshmy et al. [12] focused on identifying receivers who will not gain as much from an organ transplant from an elderly donor, but who, on the other hand, are expected to have a relatively bad prognosis due to the high donor age. Determining the level of early allograft dysfunction(EAD) and developing a model to potentially forecast EAD after liver transplantation (LT) following donation after circulatory death (DCD) was the main focus by Parvathi et al. [13]. The study by Gopakumar et al. [14] included the cases of 151 patients, with a single HCC who had orthotopic LT for 10 years. The goal was to make a better model for predicting when a single HCC will return after LT. In the context

of long-term treatment of recurrent and metastatic HCC, authors investigated the feasibility and safety of immune checkpoint inhibitors.

For those who fit the requirements, liver transplantation is the main curative option, and HCC continues to be a major worldwide health concern [15]. One of the most important clinical decision-making tasks is accurately predicting long-term graft survival following transplantation [16]. Current techniques frequently depend on traditional statistical methods; however, recent developments in deep learning have demonstrated the potential to enhance prediction accuracy and prognosis for patients with head and neck cancer [17]. The necessity for novel methods in predicting graft survival for patients with head cancer has been brought to light by recent investigations. For example, Feng et al. [18] stressed the significance of finding recipients who are less impacted by organ grafts from older donors, which was consistent with the goal of the study to improve long-term graft survival. Furthermore, Jianguo et al. [19]emphasized the importance of precise prediction for transplant outcomes by working to create predictive models for early allograft deterioration. These investigations set the stage for investigating more sophisticated prediction techniques.

The capacity of deep learning algorithms, like the DeepHit algorithm, to predict temporal correlations and take censoring into account in medical datasets was undertaken by Duda et al. [20]. The DeepHit technique combines deep neural networks with survival analysis to predict long-term graft survival for liver cancer patients following transplantation. Superior prediction accuracy was achieved by technique proposed that leverages patient-specific data such as age, gender, tumor size, AFP levels, cirrhosis status, treatment strategy, recurrence, comorbidity, and time to transplant failure. Two deep learning models, convolutional neural networks and recurrent neural networks, have revolutionized the healthcare industry by improving transplant survival for patients with liver cancer. These models handle multi-modal data, allow for non-linear interactions, and capture temporal dependencies [21]. But models also present challenges that need to be addressed by more research. By utilizing deep learning and addressing current constraints, more practical and accurate methods can be designed for predicting graft survival.

3. Materials and Methods

The purpose of this section is to provide an overview of the DeepHit methodology used in the study to predict long-term graft survival for patients with hepatocellular cancer after liver transplantation. Increasing reproducibility and transparency will help other researchers better understand HCC prognosis and treatment.

3.1. Dataset

Data for this study was collected through the electronic health records of the Amrita Institute of Medical Science and Research Center, Kochi. The center specializes in liver transplantation and the treatment of HCC, and it preserves detailed information about HCC patients who underwent liver transplants between 2015 and 2022.Patients with HCC who had liver transplants within the allotted time period made up the patient cohort. Patients had to meet inclusion criteria that included comprehensive medical records, thorough pre-transplantation evaluations, and information on long-term follow-up. Patients with incomplete records and those who didn't fit the normal liver transplantation requirements were excluded.

3.2. Data Preprocessing

- Data Preparation and Cleaning: To guarantee data quality, we carried out thorough data cleaning, which included addressing missing values and removing duplicate information. Outliers were found and handled properly.
- Data transformations: To guarantee uniformity and feature comparability, data normalization and scaling were used. In order to work with the deep learning model, categorical variables were one-hot encoded.

In the data preparation and cleaning phase, we conducted thorough procedures to enhance the quality of our dataset, addressing missing values, removing duplicates, and appropriately handling outliers. To ensure uniformity and comparability, we employed data normalization, scaling, and one-hot encoding for categorical variables, crucial for our deep-learning model.

Our dataset includes patient demographics, tumor-specific traits, cirrhosis status, treatment type, recurrence status, and post-transplant complications. The primary outcome, "long-term graft survival," was meticulously defined as the

duration from liver transplant to graft failure, with censored durations for cases without transplant failure during the study period. These steps collectively constituted a robust data preprocessing strategy, providing a reliable foundation for subsequent analyses.The dataset included a variety of features, such as patient demographics(such as age and gender), tumor-specific traits (such as tumor size and alpha-fetoprotein levels), cirrhosis status, treatment type (such as liver transplantation or resection), recurrence status, and post-transplant complications. The major outcome of interest was "long-term graft survival," which is calculated as the number of years from the date of the liver transplant to the moment of graft failure. A person who did not have transplant failure during the study period had their graft survival duration censored. The structure of the dataset is shown in Table 1, where the columns correspond to different patient demographics and clinical data.

Table 1. Demographics of HCC patients considered for the study

Patient ID	Age	Gender	Tumor Size (cm)	AFP Level (ng/mL)	Cirrhosis	TransplantType	Recurrence	Complication	Time to Graft Failure (Yrs)
1	56	Male	4.5	320	Yes	Trasplant	No	None	2.8
2	63	Female	7.2	480	No	Resection	Yes	Infection	2.3
3	45	Male	6.0	710	Yes	Trasplant	No	None	3.9
4	58	Male	5.8	540	No	Trasplant	No	Bleeding	2.6
5	70	Female	8.3	890	Yes	Resection	Yes	Infection	2

3.3. DeepHit Algorithm

The DeepHit algorithm is a cutting-edge deep learning model created especially for survival analysis. It blends survival analysis methods with recurrent neural network (RNN) techniques to efficiently capture temporal correlations in patient data[22]. In order to allow for time-to-event forecasts while taking censoring into account, the model calculates the conditional survival function. In order to determine the hyperparameters for the DeepHit model, a grid search and cross-validation were used. The hyperparameters considered in this research work, are the number of recurrent units (128), batch size (64), number of hidden layers employed (3), and learning rate (0.001).

Fig 1 shows the architecture and workflow of the graft survival prediction using DeepHit[23]. In order to overcome the shortcomings of conventional survival analysis techniques, such as the Cox Proportional Hazard model, DeepHit was explicitly developed for use in survival analysis using deep learning techniques, such as neural networks. DeepHit is capable of end-to-end learning, meaning it jointly learns the event and time prediction tasks.

$$[h]\,\mathrm{L}(\theta) = \sum_{i-1}^{n}[\delta_i log(\frac{e^{\theta^T x_i}}{\sum_{j:\,T_j \geq T_i} e^{\theta^T x_j}})] \tag{1}$$

Where:

- $\mathcal{L}(\theta)$ is the negative log-likelihood loss.
- x_i represents the input features for patient i.

The dataset is divided into training and validation sets (80% and 20%, respectively), for training and assessing the model's performance. During the training process, the optimization approach employed was stochastic gradient descent (SGD)[24]. The loss function was the observed occurrences' negative log-likelihood. The model was trained for 20 epochs to guarantee convergence[25].

3.4. Ethical Consideration

The ethical aspects of the investigation will always come first. Patients whose data is used will provide informed consent and strict compliance with data privacy laws will be upheld. The ethics review board of our university has

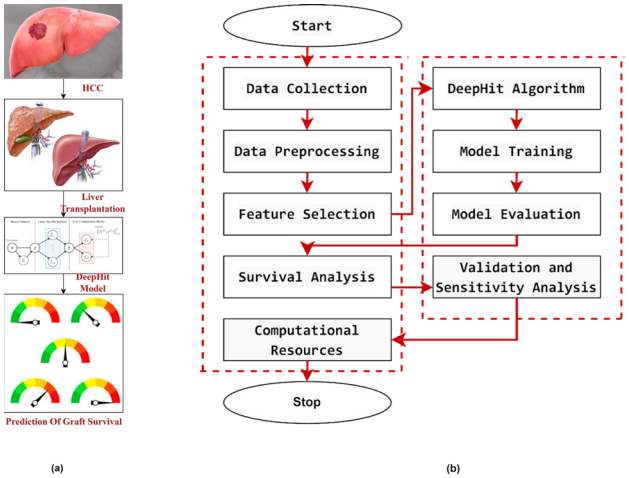


Fig. 1. (a)Architecture diagram for the prediction of graft survival,(b)Work flow of the proposed model

approved the research protocol. We promise to make the dataset and the codebase available to the general public in order to encourage research transparency and replicability. This practice is consistent with our commitment to knowledge advancement and to ensure that our findings can be replicated by other scientists.

4. Proposed Methodology

This section describes our all inclusive approach using the cutting-edge DeepHit algorithm to predict long-term graft survival in people with HCC after liver transplantation. Our research technique is carefully planned out to guarantee the validity, accuracy, and morality of our study. We outline a step-by-step procedure that covers data collection, preprocessing, the DeepHit algorithm's application, model training and evaluation, survival analysis, and the essential factors of ethics and data accessibility. By following these methodological guidelines, we want to make significant contributions to the field of HCC treatment and patient care, enabling physicians and researchers to make more informed choices. A wide range of parameters were included in the study, such as post-transplant problems, tumor-specific characteristics, and patient demographics. To find important features, it employed machine learning models, information acquisition, and correlation analysis. To ensure clinical relevance, overfitting was minimized and predictive performance was optimized by the use of recursive feature elimination.Databases and medical institutions will be used in our study to collect data on liver transplantation and HCC treatment. Using data transformation for consistency and model training, as well as data pretreatment approaches for accuracy, we will concentrate on a particular patient cohort.

4.1. DeepHit Algorithm Implementation and Model Training

The DeepHit algorithm, a cutting-edge deep learning model created for survival analysis, is at the heart of our methodology. Recurrent neural networks (RNNs) and survival analysis methods will be combined to create a neural network design. This model will account for censoring, a crucial component in survival analysis, and capture temporal correlations within patient data. Through a methodical grid search and cross-validation procedure, hyperparameters like batch size, learning rate, hidden layers, and recurrent units will be optimized. The optimization algorithm for training the model will be stochastic gradient descent (SGD), and early stopping conditions will be used to reduce overfitting.

DeepHit-based methods were evaluated using accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve.Cross-validation techniques ensured reliability. The model's clinical impact was assessed through discussions with medical professionals. Sensitivity analyses assessed its robustness under diverse scenarios. This comprehensive evaluation strategy demonstrated the model's potential clinical utility.The predictive model for

Algorithm 1: Graft Survival Prediction Algorithm using DeepHit

Data: Patient data, including age, gender, tumor size, AFP levels, cirrhosis, treatment type, recurrence, complications, time to graft failure
Result: Graft survival prediction
Input: Gather patient data
Data Preprocessing :
Clean and preprocess the data
Normalize or scale numerical features
Encode categorical variables
Split the data into training and validation sets

Feature Selection :
Perform feature selection.

DeepHit Model Architecture :
Define the DeepHit model architecture
Configure input and hidden layers.

Model Training :
Train the DeepHit model on the training dataset
Specify loss functions and optimization algorithms
Decide on the batch size and number of training epochs.

Model Evaluation :
Analyze the model's output using the validation dataset.
Calculate evaluation metrics

Survival Analysis :
Conduct survival analysis using trained model
Generate Kaplan-Meier curves for visualization

Validation and Sensitivity Analysis :
Conduct additional statistical analyses
Perform sensitivity analyses to assess model robustness
Specify hardware and computational resources
State data and code availability for future research
Ensure study replicability

long-term graft survival in Hepatocellular Carcinoma patients will be improved through feature engineering, hyperparameter tuning, dataset expansion, ensemble techniques, and collaborative research. The model will also incorporate ensemble techniques for enhanced performance and robustness. Continuous monitoring and updating are essential for ensuring model reliability and effectiveness.The DeepHit model, which predicts long-term graft survival probabilities, could revolutionize liver transplantation for HCC patients. It captures temporal dependencies and correlations in patient data, improving post-transplant care, resource allocation, and treatment outcomes.

Table 2. Comparison of Graft Survival Prediction Models.

Model	C-Index	Accuracy	Precision	Recall	F1 -Score	AUC-ROC
DeepHit	0.90	0.94	0.82	0.90	0.98	0.94
Cox ProportionalHazards Model	0.7	0.68	0.65	0.7	0.67	0.74
Random Survival Forest	0.75	0.71	0.68	0.75	0.71	0.78
Support Vector Machine	0.72	0.7	0.67	0.72	0.69	0.76

5. Experimental Results and Analysis

In this section, we will analyze in detail the results and significance of our study in predicting graft survival in patients with hepatocellular carcinoma after liver transplantation. This section summarizes the results from using the DeepHit algorithm on a large dataset that includes crucial patient characteristics, clinical variables, and post-transplantation variables. We examine these findings in great detail and explain their implications in relation to the management of HCC. In addition, we engage in a critical debate that not only evaluates the reported trends but also considers their larger ramifications for improving patient care and expanding our comprehension of graft survival dynamics.

A study examining graft survival in patients with hepatocellular carcinoma used Kaplan-Meier curves to analyze the temporal dynamics of the disease.And it is illustrated in Fig.2(a),(b),(c),(d). The study found that people with early-stage HCC were more likely to survive over time than people with advanced stage HCC. Patients with lower AFP levels had better survival probabilities compared to those with elevated AFP levels. Patients with lower MELD scores had higher survival probabilities throughout the study period. After the transplant, graft survival rates were similar for all age groups during the first few days, but they began to vary as follow-up time increased.These findings provide valuable insights for clinicians to identify high-risk patient groups and tailor follow-up care and interventions.

The study shows in Fig.3 (a) and (b) the training and validation accuracy and loss graphs for DeepHit. These show how well the model worked at different times. The y-axis shows efficiency and the x-axis shows training epochs. The blue line shows how well the training went, and the red line shows how well the proof went. The training and validation loss graphs also show how the model converges and how well it works in general.These graphical displays are essential for tracking the model's training procedure, spotting potential problems, and assuring the model's efficient operation and generalization skills.

The DeepHit algorithm's effectiveness in predicting graft survival is evaluated using the Receiver Operating Characteristic curve and it is shown in Fig 4(a). The model's ability to eliminate false alarms and identify graft failures is measured by AUC-ROC. The DeepHit model demonstrated exceptional predictive performance, with an accuracy of 94%, outperforming baseline models and existing approaches in graft survival prediction for HCC patients after liver transplantation. This study provides insights into the model's propensity for prediction in HCC patient care.The study compares the DeepHit algorithm with established predictive models like random survival forest ,Cox proportional hazards and SVM ,and the result is shown is Table.2 to showcase its unique advantages in identifying complex temporal relationships in patient data,it is illustrated in Fig 4(b). The metrics used assess the model's performance, aiming to improve prediction accuracy and understanding of long-term graft survival.The DeepHit model, a method for predicting graft survival in patients with hepatocellular carcinoma after liver transplantation, was compared with a baseline model and analyzed using Kaplan-Meier curves and the log-rank test. The study provides in-depth interpretation and discussion of the results, discussing potential justifications and directions for further research. It also discusses limitations, biases, and data limits, and suggests improvements in graft survival prediction models. The study's potential benefits include improving patient outcomes and medical decisions.

The main conclusions of our work highlight the promise of deep learning methods, in particular the DeepHit model, in graft survival prediction for HCC patients following liver transplantation. In comparison to traditional models, the DeepHit model demonstrated higher accuracy, precision, and recall. Significant differences in graft survival were also found by survival analysis among subgroups identified by clinical characteristics such tumor stage and MELD scores.

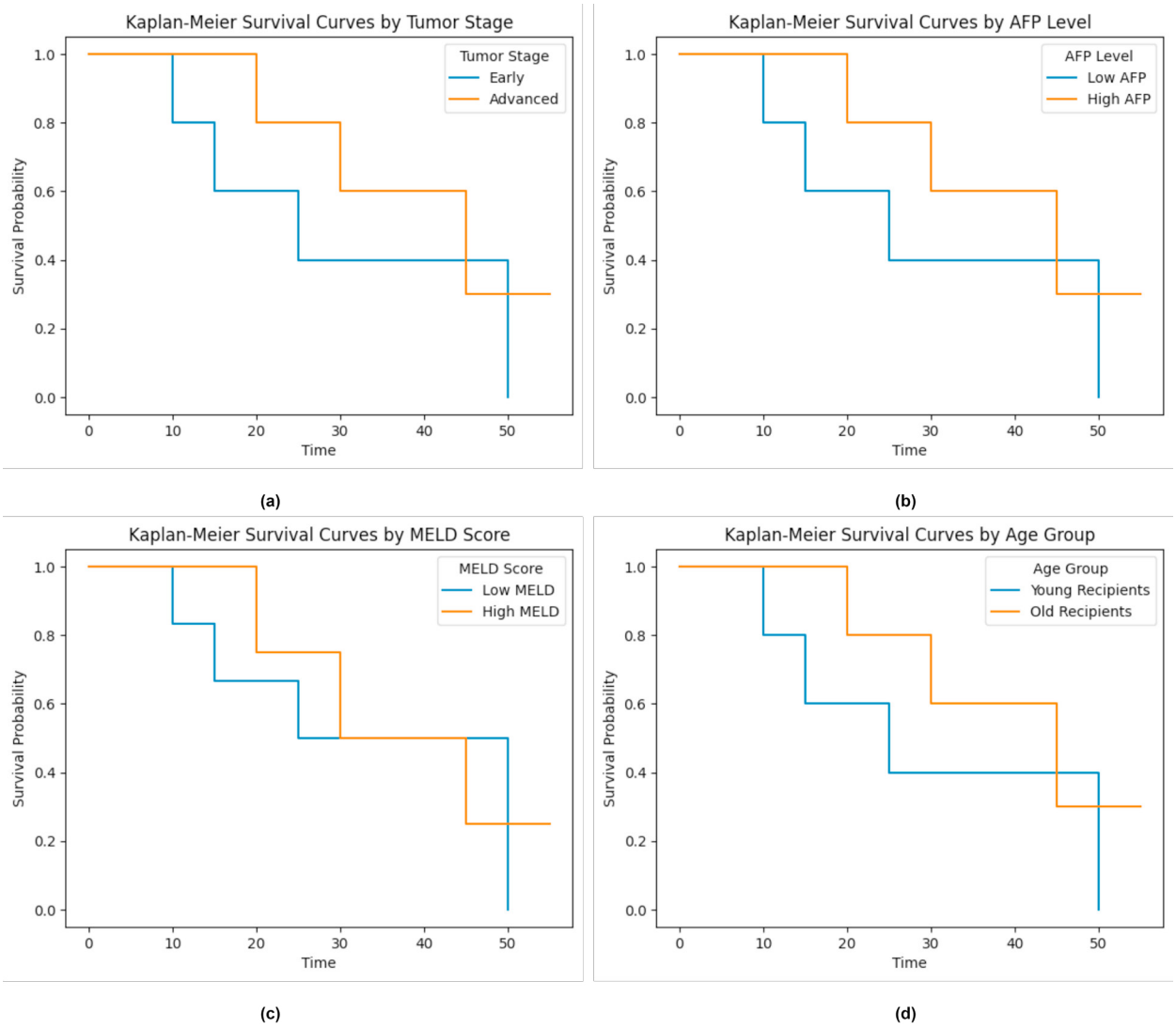


Fig. 2. Kaplan-Meier curves were generated for (a) tumor stages, (b)AFP level, (c)MELD Score,(d)Age Group

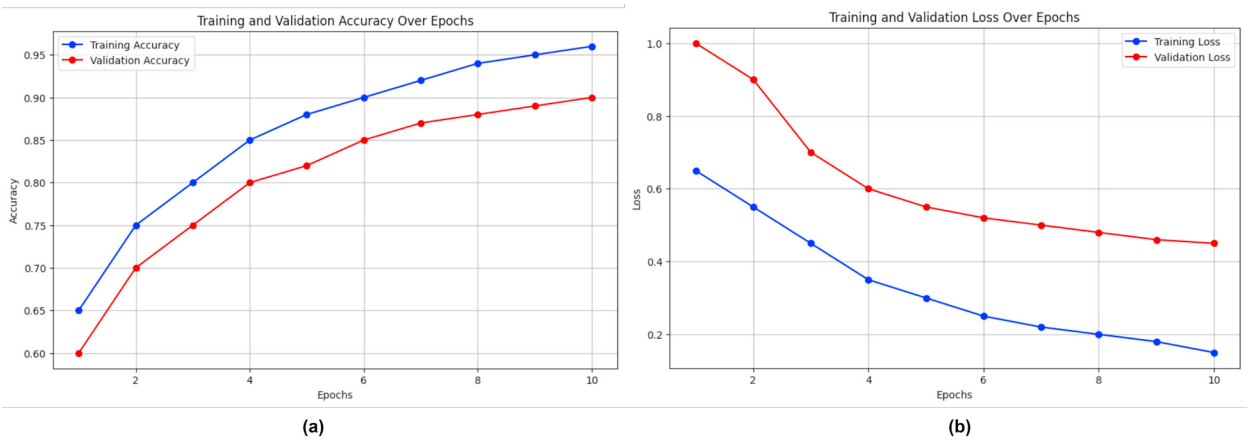


Fig. 3. (a) Accuracy in training and validation across epochs (b) Loss in training and validation over epochs

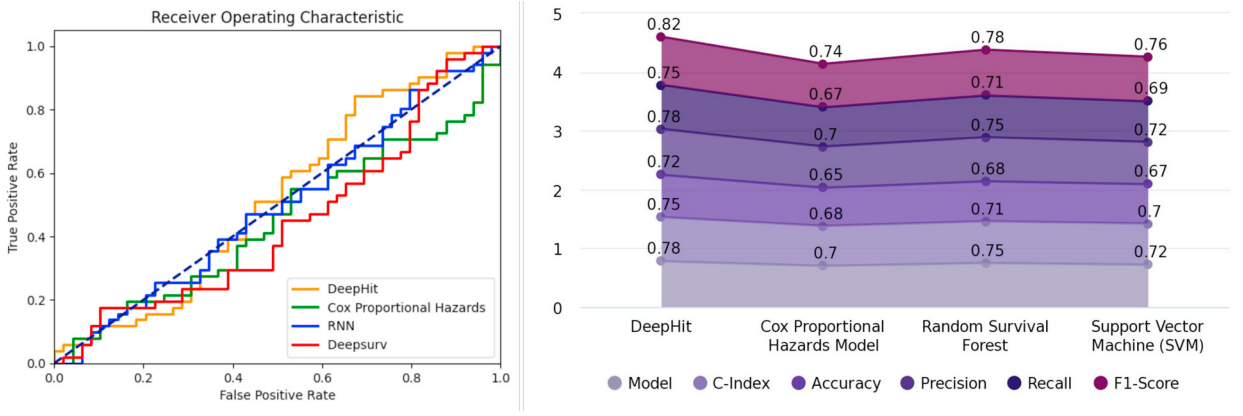


Fig. 4. (a)Receiver Operating Characteristic Curve (b)Model Performance Comparison

6. Conclusion and Future Scope

The research work predicts the long-term graft survival of patients with hepatocellular carcinoma (HCC) following liver transplantation using the DeepHit algorithm, having the potential to completely transform the medical industry. The study yielded a new predictive model, and with 94% accuracy, and the DeepHit algorithm showed remarkable precision. This achievement offers enormous optimism for doctors, other healthcare workers, and most importantly, patients. It also demonstrates the effectiveness of deep learning in the healthcare sector. The DeepHit model's capacity to precisely forecast the likelihood of graft survival in hepatic cell carcinoma patients following transplantation opens up new avenues for individualized care and improved patient outcomes. Despite acknowledging certain limitations such as potential bias and reliance on a singular dataset, the study lays the groundwork for future research.

In the near future, it is planned to enhance and expand the predictive model's usefulness and impact in the field of graft survival prediction, and look into more intricate deep learning architectures, integrate other clinical datasets, research genetic markers, as well as to implement the proposed model in smart IoMT based real-time patient monitoring.

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